MTH219 2021 S1 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

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* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

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MTH219

Timed Online Assignment – January Semester 2021

Fundamentals of Statistics and Probability

Thursday, 20 May 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Unless otherwise specified, full details of the workings for each question are to be submitted.

Answer all questions. (Total 100 marks)

Question 1

A test, in which the marks obtainable range from 0 to 100 inclusive, was taken by 30 students who had been trained in such tests, and by 20 other students who had not. A summary of the results is given in Table Q1 below.

	Mean	Standard Deviation
30 trained students	62	10
20 untrained students	52	12

Table Q1

(a) Calculate the mean and standard deviation of the combined set of 50 scores. Give your answers to two decimal places.

[Hint: you may use the result: variance, $s^2 = \frac{\sum(x-\bar{x})^2}{n} = \frac{1}{n}\sum x^2 - \bar{x}^2$]

(10 marks)

(b) One of the students was ill but took the test and scored only 1 mark. It was decided to exclude this result from the analysis.

(i) Calculate the mean and standard deviation of the remaining 49 scores. Give your answers to two decimal places.

(6 marks)

(ii) Briefly explain what would be the changes in the mode and median when compared to those of the full set of 50 scores.

(4 marks)

Question 2

In a batch of manufactured items, the proportion of defective items is p. From each batch a random sample of nine is taken and tested. If two or more items are found to be defective, the batch is rejected, otherwise it is accepted.

(a) Show that the probability that a batch is accepted is $(1-p)^8(1+8p)$.

(6 marks)

(b) It is decided to modify the checking scheme such that when one defective is found in the sample, a second sample of nine is taken and the batch rejected if this contains any defectives. With this exception, the original scheme is continued.

(i) Apply the modified scheme and derive an expression in terms of p for the probability that a batch is accepted, simplifying your answer as much as possible.

(6 marks)

(ii) Evaluate and comment on the average number of items sampled over a large number of batches, with this new sampling scheme, when $p = 0.1$. Provide your answer to 2 decimal places.

(8 marks)

Question 3

A drink dispensing machine dispenses either cups of tea or coffee. The number of cups of tea sold may be assumed to be a Poisson variable with mean of 0.5 cups per 10 minutes period, while the number of cups of coffee sold may be assumed to be a Poisson variable with mean of 1.5 cups per 10 minutes period.

(a) Compute the probability that in a given <u>half-hour period</u>, exactly 2 cups of tea and 2 cups of coffee are dispensed by the dispensing machine.

(5 marks)

(b) Calculate the probability that in a given <u>45 minutes period</u>, more than 7 drinks are dispensed by the dispensing machine.

(5 marks)

(c) In a <u>10 minutes period</u>, 3 drinks were dispensed by the dispensing machine. Calculate the probability that these were all cups of coffee.

(5 marks)

Question 4

During peak hours, the number of bus "74" arriving at the bus stop outside SUSS campus is a Poisson variable with rate of 9 buses per hour.

(a) Calculate the probability that you would need to wait more than 20 minutes for the bus "74" if one bus "74" just departed on your arrival at the bus stop. Show full details of your workings.

(6 marks)

(b) Briefly explain if you would be expected to wait longer at the bus stop if the bus "74" has just departed when you arrive at the bus stop.

(4 marks)

Question 5

In a city, the mean height of married man is 180 cm with a standard deviation of 4 cm, the mean height of married women is 175 cm with a standard deviation of 3 cm. Assume that the choice of partner in marriage is not influenced by height consideration. A couple is selected at random. Showing all details of your workings for the following computations.

(a) Compute the probability that

(i) both of them are taller than 177.5cm;

(4 marks)

(ii) the husband is taller by less than 5cm;

(4 marks)

(iii) their height difference is less than 5cm.

(4 marks)

(b) 7 more married couples are selected from the city. Compute the probability that at least 2 of the selected couples have height differences of more than 5cm.

(8 marks)

Question 6

The life in days, X, of an insect is such that $\log_{10} X$ is normally distributed with mean of 2 days and standard deviation of 0.2 days.

(a) Calculate the probability that an insect will have a life of

(i) more than 200 days;

(3 marks)

(ii) between 50 and 150 days inclusive.

(4 marks)

(b) Two insects have life expectancies of T_1 and T_2, and $H = \frac{T_1}{T_2}$.

(i) Show and describe the distribution type of $\log_{10} H$, if their life expectancies are independent.

(4 marks)

(ii) If one insect has life expectancy 1.8 times that of the other, is this difference significant?

(4 marks)

----- END OF PAPER -----

1. (a)

let X represent the 30 trained students
let Y represent the 20 untrained students
let Z represent the combined 50 students

$E(X) = 62$, $var(X) = 10^2 = 100$
$E(Y) = 52$, $var(Y) = 12^2 = 144$

$$\sum_{i=1}^{50} z_i = \sum_{i=1}^{30} x_i + \sum_{i=1}^{20} y_i$$
$$= 30E(X) + 20E(Y)$$
$$= 2900$$
$$E(Z) = \frac{1}{50}\sum_{i=1}^{50} z_i = \frac{2900}{50} = 58$$

$$var(Z) = E(Z^2) - [E(Z)]^2$$
$$= \frac{1}{50}\sum_{i=1}^{50} z_i^2 - \bar{z}^2$$

$$\sum_{i=1}^{50} z_i^2 = \sum_{i=1}^{30} x_i^2 + \sum_{i=1}^{20} y_i$$

$$var(X) = E(X^2) - [E(X)]^2$$
$$10^2 = \frac{1}{30}\sum_{i=1}^{30} x_i^2 - \bar{x}^2$$
$$\frac{1}{30}\sum_{i=1}^{30} x_i^2 = 10^2 + \bar{x}^2 = 3944$$
$$\sum_{i=1}^{30} x_i^2 = 30 \times 3944 = 118320$$

$$var(Y) = E(Y^2) - [E(Y)]^2$$
$$12^2 = \frac{1}{20}\sum_{i=1}^{20} y_i^2 - \bar{y}^2$$
$$\frac{1}{20}\sum_{i=1}^{20} y_i^2 = 12^2 + \bar{y}^2 = 2848$$
$$\sum_{i=1}^{20} y_i^2 = 56960$$

$$\sum_{i=1}^{50} z_i^2 = \sum_{i=1}^{30} x_i^2 + \sum_{i=1}^{20} y_i^2$$
$$= 175280$$

$$var(Z) = \frac{1}{50}\sum_{i=1}^{50} z_i^2 - \bar{z}^2$$
$$= \frac{175280}{50} - 58^2$$
$$= 141.6$$

$$\therefore sd(Z) = \sqrt{var(Z)}$$
$$= 11.89958$$
$$\simeq 11.90$$

$$\therefore E(Z) = 58$$

1(b) let $z_{50} = 1$

$$\text{new } E(Z) = \frac{1}{49}\sum_{i=1}^{49} z_i$$
$$= \frac{1}{49}\left[\sum_{i=1}^{50} z_i - z_{50}\right]$$
$$= \frac{2899}{49}$$
$$\simeq 59.16$$

$$\sum_{i=1}^{49} z_i^2 = \sum_{i=1}^{50} z_i^2 - z_{50}^2$$
$$= 175279$$

$$\text{new } var(Z) = \frac{1}{49}\sum_{i=1}^{49} z_i^2 - [E(Z)]^2$$
$$= \frac{175279}{49} - \left(\frac{2899}{49}\right)^2$$
$$= 76.218724$$

$$\text{new } sd(Z) = \sqrt{var(Z)}$$
$$= 8.730306$$

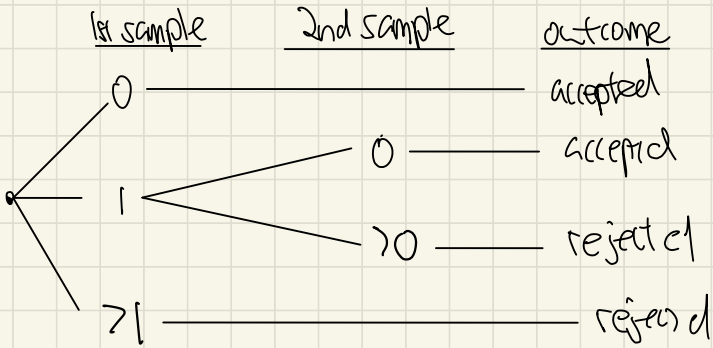
By Central Limit Theorem, both sets of scores are approximated by normal distribution, therefore the mean is also the mode and median. Since the mean increased from 58 to 59.16, the mode and median also increased.

2(a) let X be the no. of defective items in a sample of 9.

$X \sim Bin(9, p)$

$$
\begin{aligned}
P(\text{batch accepted}) &= 1 - P(\text{Batch rejected}) \\
&= 1 - P(X \geq 2) \\
&= P(X=0) + P(X=1) \\
&= \binom{9}{0} p^0 (1-p)^9 + \binom{9}{1} p (1-p)^8 \\
&= (1-p)^9 + 9p(1-p)^8 \\
&= (1-p)^8 [(1-p) + 9p] \\
&= (1-p)^8 (1+8p)
\end{aligned}
$$

(b)



$$
\begin{aligned}
P(\text{batch accepted}) &= P(X=0) + P(X=1)\,P(X=0) \\
&= \binom{9}{0} p^0 (1-p)^9 + \binom{9}{1} p (1-p)^8 \cdot \binom{9}{0} p^0 (1-p)^9 \\
&= (1-p)^9 (1 + 9p(1-p)^8)
\end{aligned}
$$

average number of items surveyed per batch

$$
\begin{aligned}
&= 9[P(X=0) + P(X>1)] + 18\,P(X=1) \\
&= 9[P(X=0) + 1 - P(X \leq 1)] + 18\,P(X=1) \\
&= 9 \times [\text{dbinom}(0, 9, p) + 1 - \text{pbinom}(1, 9, p)) + 18 \times \text{dbinom}(1, 9, p) \quad \text{where } p = 0.1 \\
&= 12.48678 \simeq 12.49
\end{aligned}
$$

8 let X_T be no. of tea cups dispensed during a T-min interval
let Y_T be no. of coffee cups dispensed during a T-min interval
let Z_T be no. of drinks dispensed during a T-min interval

$$Z_T = X_T + Y_T$$
$$X_{10} \sim Po(0.5)$$
$$Y_{10} \sim Po(1.5)$$

(a) $X_{30} \sim Po(1.5)$, $Y_{30} \sim Po(4.5)$

$$\begin{aligned} P(X_{30}=2 \text{ and } Y_{30}=2) &= P(X_{30}=2) \cdot P(Y_{30}=2) \\ &= \text{dpois}(2, 1.5) \times \text{dpois}(2, 4.5) \\ &= 0.02823454 \\ &\simeq 0.0282 \end{aligned}$$

(b) $Z_{10} \sim Po(2.0)$
$Z_{45} \sim Po(9)$.

$$\begin{aligned} P(Z_{45} > 7) &= 1 - P(Z_{43} \leq 7) \\ &= 1 - \text{ppois}(7, 9) \\ &= 0.676103 \\ &\simeq 0.676 \end{aligned}$$

(c) P(all drinks are coffee | 3 drinks dispensed in 10 min)

$$= \frac{P(X_{10}=0) \cdot P(Y_{10}=3)}{P(Z_{10}=3)}$$

$$= \frac{\text{dpois}(0, 0.5) \times \text{dpois}(3, 1.5)}{\text{dpois}(3, 2)}$$

$$= 0.421875$$
$$\simeq 0.422$$

4(a) let X be the number of bus 74 arriving in 1 h
let T be the time interval between 2 successive buses 74

$$X \sim Po(9 \text{ per } h) \rightarrow T \sim Exp(9 \text{ per } h)$$

$$\begin{aligned} p(T > 20\text{min}) &= p(T > \tfrac{1}{3}h) \\ &= 1 - p(T \leq \tfrac{1}{3}h) \\ &= 1 - \text{pexp}(\tfrac{1}{3}, 9) \\ &= 0.04978707 \\ &\simeq 0.0498 \end{aligned}$$

(b) The exponential distribution is memoryless, so it doesn't matter how long you previously had waited. You should not expect to wait longer if the bus 74 just departed.

5 let M be height of a married man
let W be height of a married woman
$M \sim N(180, 4^2)$ $W \sim N(175, 3^2)$

(a) i. P(both of them taller than 177.5cm)
$= P(M > 177.5) \cdot P(W > 177.5)$
$= [1 - P(M \leq 177.5)][1 - P(W \leq 177.5)]$
$= (1 - \text{pnorm}(177.5, 180, 4)) * (1 - \text{pnorm}(177.5, 175, 3))$
$= 0.148512$
$\simeq 0.149$

ii. $E(M-W) = E(M) - E(W) = 5$
$Var(M-W) = Var(M) + Var(W) = 5^2$
$M-W \sim N(5, 5^2)$

P(Husband taller by less than 5cm)
$= P(0 \leq M-W \leq 5)$
$= \text{pnorm}(5,5,5) - \text{pnorm}(0,5,5)$
$= 0.3413447$
$\simeq 0.341$

iii. P(height difference less than 5cm)
$= P(-5 \leq M-W \leq 5)$
$= \text{pnorm}(5,5,5) - \text{pnorm}(-5,5,5)$
$= 0.4772499$
$\simeq 0.477$

(b) let X be number of couples with height difference > 5cm
$X \sim Bin(6, p)$ where $p = 1 - P(-5 \leq M-W \leq 5) = 0.5227501$
$P(X \geq 2) = 1 - P(X \leq 1)$
$= 1 - \text{pbinom}(1, 6, p)$
$= 0.9737254$
$\simeq 0.973$

6

$$\log_{10} X \sim N(2, 0.2)$$

(a) i. $P(X > 200) = P(\log_{10} X > \log_{10} 200)$

$= 1 - P(\log_{10} X \leq \log_{10} 200)$

$= 1 - \text{pnorm}(\log10(200), 2, 0.2)$

$= 0.0661427 6$

$\simeq 0.066$

ii. $P(50 \leq X \leq 150) = P(\log_{10} 50 \leq \log_{10} X \leq \log_{10} 150)$

$= P(X \leq \log_{10} 150) - P(X \leq \log_{10} 50)$

$= \text{pnorm}(\log10(150), 2, 0.2) - \text{pnorm}(\log10(50), 2, 0.2)$

$= 0.7445512$

$\simeq 0.745$

(b)

i. $\log_{10} H = \log_{10} \frac{T_1}{T_2} = \log_{10} T_1 - \log_{10} T_2$

$E(\log_{10} H) = E(\log_{10} T_1) - E(\log_{10} T_2) = 2 - 2 = 0$

$\text{var}(\log_{10} H) = \text{var}(\log_{10} T_1) + \text{var}(\log_{10} T_2) = 0.2^2 + 0.2^2 = 0.08$

$\therefore \log_{10} H \sim N(0, 0.08)$

ii.

consider 95% confidence level for upper 1-tail test

$\log_{10} X_{critical} = \text{qnorm}(0.95, 2, \text{sqrt}(0.08))$

$= 2.465235$

since 1.8 isn't in the rejection region, then the difference is not significant.